

Qatar-1b

R-0473

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_180709 · created 2026-07-20T02:16:13+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458308.7601 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.057
Transit depth (Rp/Rs)^2	2.81 +/- 1.9 [%]
Orbital Inclination (inc)	80.87 +/- 2.88
Airmass coefficient 1 (a1)	0.26 +/- 0.17
Airmass coefficient 2 (a2)	1.16 +/- 1.06
Scatter in the residuals of the lightcurve fit is	65.29 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	3.0
Transit Duration (day)	0.037 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.168 $\pm$ 0.057 vs 0.1463 (deviation 0.38 σ)
Duration fit vs published	0.037 d vs 0.0692 d (deviation 1.11 σ)
Mid-time O-C	1.8 min
Depth SNR	2.9
Residual scatter	65.29 %

Light curve

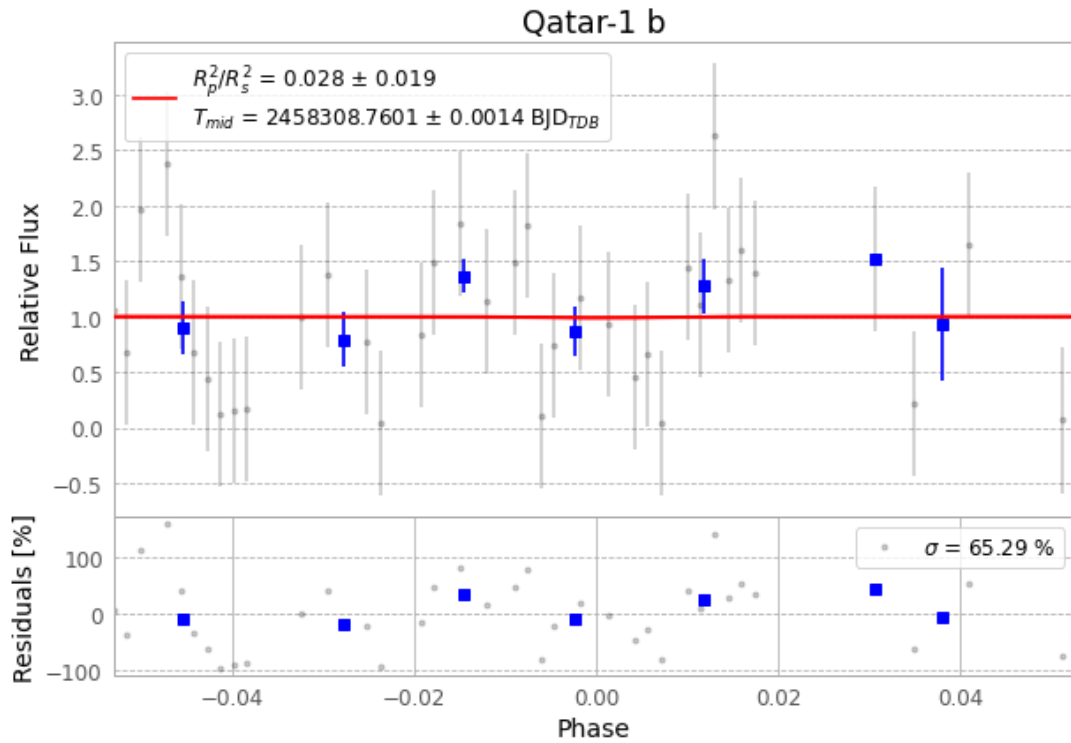


Figure 1 — FinalLightCurve_Qatar-1 b_2018-07-08.png (SHA-256 413359cad6bda02d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [115, 393], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d4e0847eef58b67f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

73 FITS frames (48 MB), Qatar-1180709042412.FITS → Qatar-1180709080012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-180709.log (f132ca7fd35f...), FinalLightCurve_Qatar-1 b_2018-07-08.png (413359cad6bd...), FinalLightCurve_Qatar-1 b_2018-07-08.csv (0df0ba8c9e87...)

Compute resources

Wall time 190.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 02:16Z.